

Evaluation of Acceptable Charging Current of Power Li-Ion Batteries Based on Polarization Characteristics

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Abstract—Battery polarization voltage characteristics of power Li-ion batteries are investigated at different charging rates. The polarization voltage is small and relatively stable in the middle state of charge (SOC) range, whereas at both ends of SOC, the polarization voltage is enlarged and varies rapidly. Meanwhile, there is an approximate linear relationship between polarization voltage and charging rate. Taking polarization time constant into account, an acceptable charging current curve is developed, which prevents charging current from being out of range. This curve takes polarization voltage and charge cutoff voltage as constraints to describe acceptable charging current value at different SOC. Based on this curve, the battery can be charged from 20% SOC to 80% SOC within 33 min. The average polarization and the maximum temperature rise are approximately equivalent to those of 0.5C constant current–constant voltage charging method. The experimental results demonstrate that the proposed charging mode can balance charging speed, charging polarization, and temperature rise.

Index Terms—Acceptable charging current, electric vehicles (EVs), polarization characteristics, power Li-ion batteries.

I. INTRODUCTION

LI-ION BATTERIES with high energy density, good cycle-life performance, and low self-discharge rate are getting more widely used in electric vehicles (EVs) and energy storage systems [1], [2]. However, the Li-ion battery charging strategy has become the bottleneck of their application, due to the slow charging speed and uncertainty effects on battery life. The charging process is closely related to battery state of health [3], [4]. The Li-ion power battery used in EVs is made up of many cells layer-by-layer, and its capacity is large, which is usually tens of ampere hours. In addition, the Li-ion power battery charging optimization technology is one of the difficult hotspots.

Li-ion power batteries are widely charged by using a C/3-rate constant current–constant voltage (CC–CV) charging method. First, the battery is charged at a CC until the battery voltage rises to the cutoff voltage. Then, the battery is charged under CV by continuously decreasing the current until the current reaches a preset value. This method is simple and easy to

control, but the charging process is slow and generally needs more than 3 h. Nowadays, lots of efforts [5]–[9] have been made to improve battery charging performance from the views of electrochemical reaction mechanism analysis and charging control algorithm. In [5], cell component variables are studied to check their effects on charging time, and it is found out that the thickness of the cathode has the greatest impact on the charging time. Hence, decreasing the thickness of the cathode is useful for reducing the charging time. These studies try to improve battery charging performance, according to the electrochemical kinetics and thermodynamics principle. However, the parameters are difficult to be measured, and the practical application is not promising. In [6], a boost-charging method is proposed. At the beginning, the battery is charged at CV to increase the charging rate. Then, a standard CC–CV period is applied. The boost-charging method can also be denoted as CV–CC–CV mode. In [7], the Li-ion battery is charged with the optimal rapid charging current pattern, which is determined by experimental results. This Taguchi approach controlling method can charge a battery to 75% capacity in 40 min. Reference [8] establishes a new charging method, whereby sinusoidal ripple current charging strategy is applied. However, the charging performance with the method is affected by minimum-ac-impedance frequency $f_{Z\min}$, which is influenced by test condition such as the circuit connection. $f_{Z\min}$ may fluctuate severely even if the battery is at the same state of charge (SOC). In [9], a pulsed current waveform with restrictions on the lithium concentration building up at the interface is proposed. These methods optimize charging time and charging efficiency on the basis of experience, but do not take the interior characteristic of the battery into account.

In 1972, Joseph A. Mas put forward three charging laws for lead–acid batteries, taking gas evolution as control condition. The laws suggest that adopting pulse charging methods can significantly improve the charging acceptance ability of lead–acid batteries. Based on the aforementioned principle, researchers have proposed pulse charging methods for lead–acid batteries. Electrochemical reaction mechanism and characteristics of Li-ion power batteries are different from lead–acid batteries [10]–[12]. In particular, the impact of periodic current pulses on Li-ion battery performance does not seem to induce any positive effects on the cell materials or electrochemical processes [13]. It is unsuitable to copy the Mas law to deal with Li-ion batteries charging. Similar to the gas evolution of lead–acid batteries, polarization of Li-ion batteries is internally generated during electrochemical reactions. The polarization influencing the charging performance externally is closely related to the

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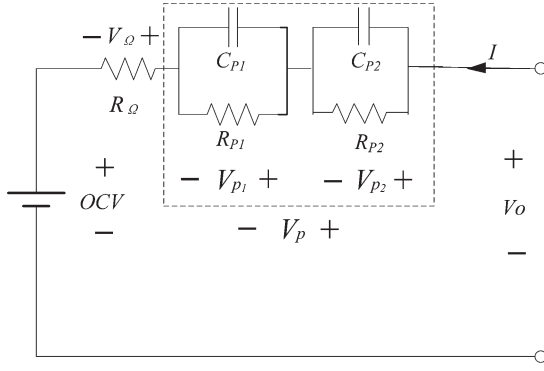


Fig. 1. Second-order equivalent circuit model of the battery.

battery life and charging efficiency. Large polarization not only renders the battery undercharged but also results in active material loss, thus affecting the battery life.

In this paper, LiFePO₄ (LFP) batteries with nominal capacity of 60 Ah will be taken as the research object. A polarization-based charging boundary curve is proposed. Based on this curve, through changing the charging current, the goal to improve charging speed and reduce polarization of the battery at the same time can be achieved. The simulating calculation error of the acceptable charging current is within 10%. The comparisons between the optimized method and the CC–CV charging method suggest that the battery can be charged from 20% SOC to 80% SOC about 33 min and the maximum temperature rise is 7 K. The instrumentations used to perform the following experiments in this paper are an Arbin Instrument BT2000 battery testing system, the thermostat, and the Fluke thermal infrared imager. The maximum voltage and current of the Arbin testing system are 5 V and 100 A, respectively. The battery is placed in the thermostat. Its temperature measuring range is 233.15–373.15 K.

II. POLARIZATION VOLTAGE CHARACTERISTICS ANALYSIS

A. Polarization Voltage Model and Test Methods

A Li-ion battery equivalent circuit model is suitable for electric system applications [14]–[16], which is illustrated in Fig. 1, where R_Ω is the ohmic resistance, R_{P1} is the charge transfer resistance, and C_{P1} is the electric double-layer capacitance of the electrode interface. R_{P2} is the diffusion resistance of charge in the electrodes and the electrolyte, and C_{P2} is the concentration diffusion capacitance. V_O is the battery terminal voltage, I is the charging current, and OCV represents the open-circuit voltage. V_P is the polarization voltage of the battery.

Polarization voltage can be calculated from (1), according to the aforementioned model

$$V_P(SOC) = V_O - OCV(SOC) - IR_\Omega. \quad (1)$$

Battery terminal voltage and charging current can be easily measured in real time. The polarization voltage is small at a very low current. Values of charging and discharging polarization voltage are approximately equal to each other, but the signs are opposite in the steady state. Ohmic resistance voltages are the same. The OCV curve can be therefore obtained by

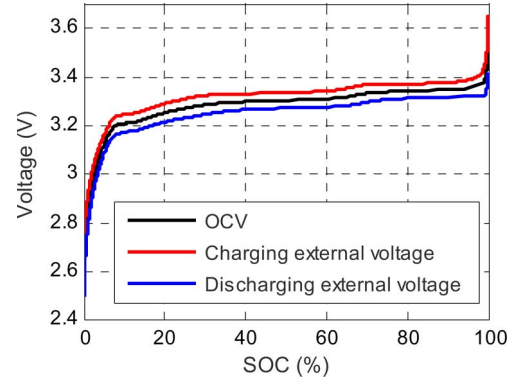


Fig. 2. Curve of OCV versus SOC.

calculating the average values of a fully charging curve, and a thorough discharge curve at low currents such as 0.05C. The relational expressions of the model are as follows:

$$\begin{aligned} V_O^{\text{ch}} &= OCV + V_\Omega + V_P \\ V_O^{\text{dis}} &= OCV - V_\Omega - V_P. \end{aligned} \quad (2)$$

OCV can be calculated by

$$OCV = (V_O^{\text{ch}} + V_O^{\text{dis}}) / 2. \quad (3)$$

Fig. 2 shows the calculated OCV – SOC curve. In [17] and [18], the authors estimate the SOC based on the Thevenin model or an improved Thevenin model. The offline identification results of the ohmic resistance parameters illustrate that the ohmic resistance at various SOC is nearly constant, independent of the battery load current. Therefore, the ohmic resistance at a certain temperature is regarded as a constant in this paper. The ohmic resistance is the ratio of voltage changes to current changes, which can be described as

$$R_\Omega = \Delta V_O / \Delta I. \quad (4)$$

According to (1), a V_P – SOC curve is available.

B. Polarization Voltage Characteristics as a Function of SOC

Charging polarization voltage profiles at various CC–CV charging rates are shown in Fig. 3. It is indicated in Fig. 3 that the initial polarization voltage is large and then it declines slowly, with the local minimum value at around 10% SOC point. The polarization voltage then becomes relatively stable, but its amplitude still fluctuates slightly. When SOC reaches about 80%, the polarization voltage starts to increase sharply.

The polarization voltage variation within the whole SOC range is closely related to the internal charging reaction process. As shown in Fig. 4, the external charger drives the flow of electrons to interact with electrode materials. Before charging, positive cathode material is in the Li-rich phase LFP. After absorbing enough energy, Li^+ becomes active exceeding the limit of activation barrier. Li^+ will move along with the direction which requires a minimum energy of the olivine-like structure. Reference [19] reports that the nucleation of LFP phase occurs

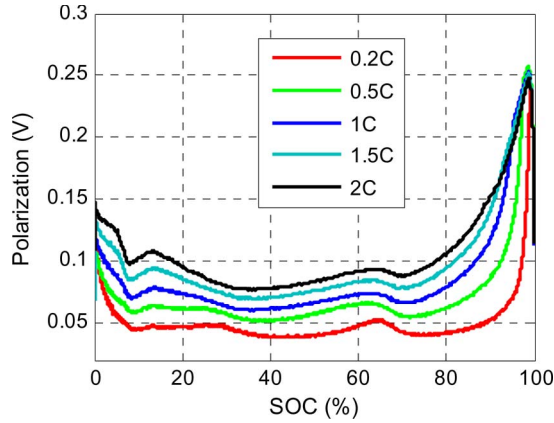


Fig. 3. Polarization voltage profiles versus SOC at various charge rates.

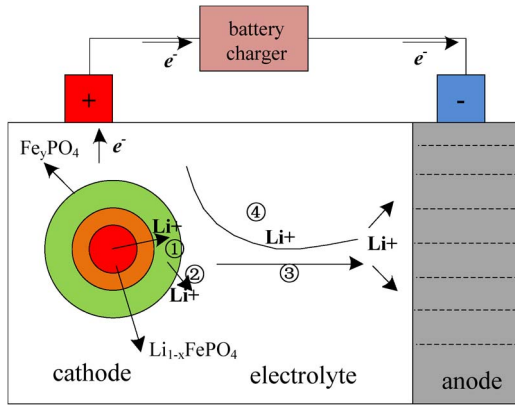


Fig. 4. Charging process diagram. ① represents Li-ion transfer in the solid-state phase, ② represents Li-ions move into the electrolyte from electrode material, ③ shows Li-ions' diffusion, and ④ is the Li-ion concentration gradients in the electrolyte.

first on the surface of LiFePO_4 when Li^+ is removing. This process needs relatively more energy. Hence, in the initial SOC region, polarization voltage is higher. Once the FP phase is formed on particle surface, during the following charging process, the sequential path of Li-ions extracting from particles is first through the core of Li-rich LFP phase, then the two-phase coexistent interface, and finally the shell of Li-poor FP phase. The Li-ions' flow is then driven by concentration potential in electrolyte. Finally, Li-ions are inserted into the graphite anode. Because FP nucleation has been finished in the low SOC region, the whole process does not need excess energy to overcome the interface energy barrier. The polarization voltage is therefore relatively low and stable in the middle SOC range.

As charging continues, the shell is growing, the core is consuming, and the two-phase coexistent interface moves toward interior. The charging process completes when the cathode material particles become the single FP phase. In the last charging stage, the travel path of Li-ions through Li-poor phase is long; in addition, with Li-ion concentration increasing in the negative materials, it is difficult for Li-ions to insert. The polarization voltage therefore shows obvious growth in the high SOC region.

The correlation between polarization voltage and charging rate is illustrated in Fig. 5, which can be divided into three parts. One is nonlinear regions containing 0%–5% SOC and

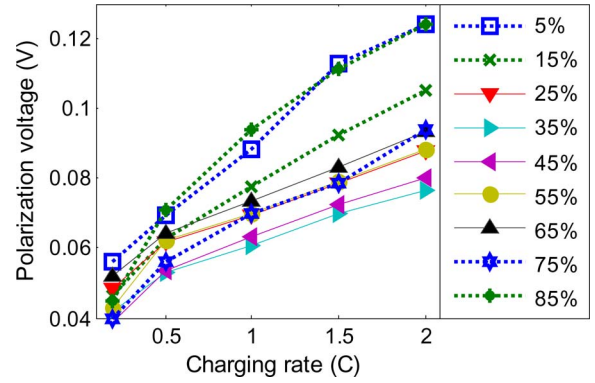


Fig. 5. Polarization voltage versus charging rate at various SOC.

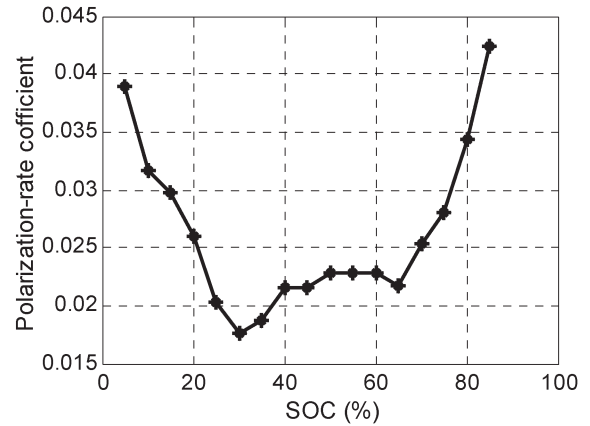


Fig. 6. PRC versus SOC.

85%–100% SOC. Another is approximately linear regions including 5%–25% SOC and 65%–85% SOC. The slopes are different between them. The third is also linear regions, and the slopes are nearly the same between them. The slopes at 5% and 85% SOC are bigger than the other SOC points, which suggests that the polarization at these two SOC points are sensitive to the charging rate. The polarization voltage will be significantly enlarged if the charging rate increases, and vice versa.

The polarization-rate coefficient (PRC) κ is defined in the study, which refers to the polarization voltage increment while charging rate is raised per 1C. It describes the degree to which charging current affects polarization voltage. In 5%–85% SOC range, the linearization slopes variably. At 5% and 85% SOC, the PRC is greater than others.

Linear fitting is conducted on the curves in Fig. 5 to calculate κ . Its relationship with SOC is shown in Fig. 6. It is shown in Fig. 6 that κ is smaller in the middle SOC region than that in both ends of the whole range. In the early and last charging stages, Li-ions gather in one electrode. Li-ion concentration is relatively high, and the diffusion coefficient is smaller. The solid-state phase diffusion polarization dominates the total polarization. Polarization rate performance is bad. In the middle SOC region of charging, Li-ions distribute well in both electrodes. The solid-state phase diffusion polarization and concentration polarization play a major role. The diffusion coefficient is larger, and Li-ions have enough free space to move easily. The polarization is relatively small.

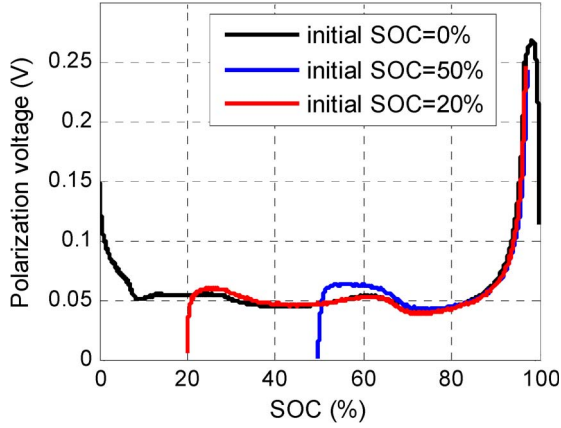


Fig. 7. Charge polarization characteristics at different initial SOC.

A series of experiments are conducted to investigate the effects of initial SOC on PRC. The procedure can be summarized as follows: the battery is first charged from fully discharged state ($SOC_0 = 0\%$) at the current of $0.5C$ until the terminal voltage is up to 3.65 V. The battery is then kept in the open-circuit state for 1 h. The battery is discharged at the current of $0.5C$ until 80% nominal capacity is consumed, and then, it is kept at open-circuit state for 1 h to eliminate the polarization of the battery. The battery is charged from initial $SOC_0 = 20\%$ at the same current until the charge limit voltage is obtained. Similarly, having been kept at open-circuit state for 1 h, the battery is discharged at $0.5C$ until 50% of nominal capacity is consumed. Leaving the battery at open-circuit state for 1 h, the battery is then charged from $SOC_0 = 50\%$ until up to the charge limit voltage. The charge polarization characteristics at different initial SOC are reported in Fig. 7. It is shown in Fig. 7 that the dynamic overshoot values of the polarization voltage are affected by the initial SOC value, which is higher at $SOC_0 = 50\%$ than that at $SOC_0 = 20\%$. However, the steady value of the polarization voltage at various initial SOC is approximately the same, which suggests that the steady value of the polarization voltage is independent of initial SOC.

Based on the aforementioned comprehensive analysis, the polarization voltage can be described by

$$\int_0^t \frac{idt}{Q} = SOC$$

$$V_P(i) = \kappa(SOC)i + b. \quad (5)$$

Q is the battery measured capacity charged at the current of $1/3C$. The measured charging capacity of CC-CV at $1/3C$ is used as the nominal capacity. In this paper, b is a linear fitting constant, which is calculated from the data shown in Fig. 5.

During battery resting time, charging current is zero. The electrical behavior of the circuit is expressed in

$$\begin{cases} V_{P1}(t)/R_{P1} = i(t) - C_{P1}dV_{P1}(t)/dt \\ V_{P2}(t)/R_{P2} = i(t) - C_{P2}dV_{P2}(t)/dt \\ V_O(t) = OCV(t) + i(t)R_O + V_{P1}(t) + V_{P2}(t). \end{cases} \quad (6)$$

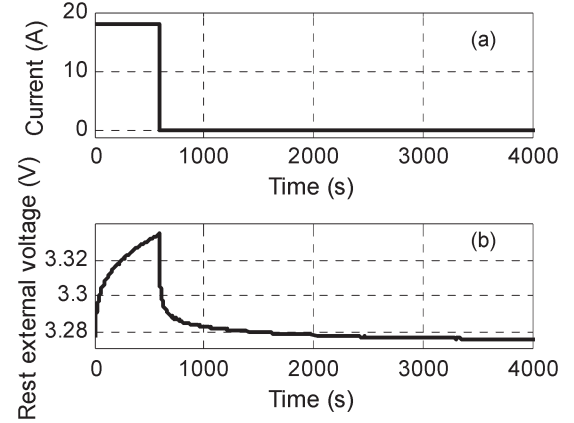


Fig. 8. Voltage and current curves change profiles.

TABLE I
POLARIZATION TIME CONSTANT AT VARIOUS SOC POINTS

SOC	τ_1/s	τ_2/s
15%	874.58	41.89
25%	983.15	34.87
35%	1260.36	40.08
45%	1074.31	43.62
55%	910.05	45.39
65%	892.37	43.97
75%	991.72	42.87
85%	787.61	49.04

The solution of (6) can be given by

$$\begin{cases} V_{P1}(t) = V_{P1}(0)e^{-\frac{t}{R_{P1}C_{P1}}} + \frac{e^{-\frac{t}{R_{P1}C_{P1}}}}{C_{P1}} \int_0^t e^{\frac{t}{R_{P1}C_{P1}}} i(t) dt \\ V_{P2}(t) = V_{P2}(0)e^{-\frac{t}{R_{P2}C_{P2}}} + \frac{e^{-\frac{t}{R_{P2}C_{P2}}}}{C_{P2}} \int_0^t e^{\frac{t}{R_{P2}C_{P2}}} i(t) dt \\ V_P(t) = V_{P1}(t) + V_{P2}(t) \\ V_O(t) = OCV(t) + i(t)R_O + V_P(t). \end{cases} \quad (7)$$

It can be observed from (7) that, when the battery stops charging suddenly, polarization voltage will decline as an exponential function, containing two time constants τ_1 and τ_2 . The polarization charging time constant (PCTC) can be determined through experiments directly. It is considered that the PCTC is approximately equal to the polarization standing time constant (PSTC). The voltage variation of a battery charged to a certain SOC is monitored for 2 h. It is found that the change of terminal voltage is around 1 mV from 3600 to 7200 s (the voltage ranging from 3.767 to 3.768 V), suggesting that the terminal voltage is getting steady after 1 h. Therefore, the terminal voltage change for 1 h is approximately regarded as the steady value of the polarization voltage. Fig. 8 shows the changing polarization voltage curve of the battery charged to 15% SOC with CC and then resting for 1 h.

Polarization voltage is obtained by using least squares fitting to identify the time constant. The PSTC of 15% – 85% SOC at every 10% SOC interval is shown in Table I. The results suggest that two time constants are of different orders of magnitude: one is 10^3 s, and the other is 10 s.

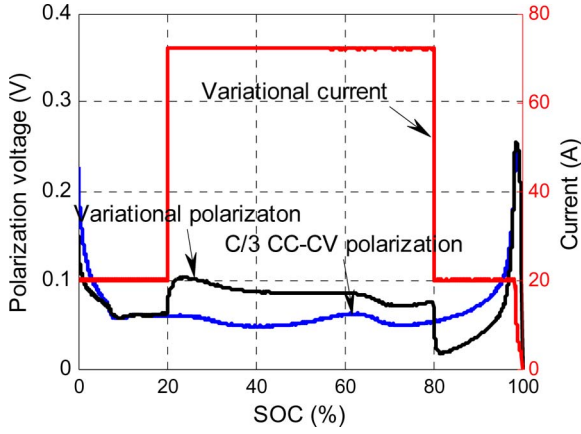


Fig. 9. Polarization step response experiment results.

C. Dynamic Polarization Voltage Characteristics

Fig. 9 is the comparison of polarization voltages under variable current charging and 1/3C CC–CV charging. The variable current charging profile is: charge the battery at 1/3C at the starting stage to 20% SOC, then charge at 1.2C during the middle stage of 20%–80% SOC, and 1/3C charge in the last stage. It is found in Fig. 9 that the battery polarization voltage overshoot appears as the charging current is increasing. On the contrast, the polarization voltage hysteresis can be observed while the charge current decreases. It is noticeable that the hysteresis in this study is resulted from current change, and disappears as the battery polarization is getting steady. The overshoot amplitudes at 20% SOC and 80% SOC points are not the same, since in the current increasing and decreasing process, reaction kinetics is not symmetrical, which results in charging and discharging polarization asymmetric [20], [21].

The whole process of polarization includes at least three aspects: electronic and ionic conduction inside the electrode, charge transfer reaction at the electrode and electrolyte interface, and Li-ion transport process such as concentration gradient-driven diffusion. They can be divided into two types. One includes the first and second fast process, and the other includes the third slow process [19]. The number of the electrons flowing through the electrode per second changes rapidly when charging current changes, but due to the time constant, these three processes cannot quickly respond. The polarization hysteresis effect is produced. The overshoot effect is a result of the excess energy cost in the initial period. However, according to (7), the over-damping response has no overshoot effect. This phenomenon indicates that the mechanism of overshoot effect cannot be described by the equivalent circuit model shown in Fig. 1.

III. PACC AND TEST ANALYSIS

The polarization acceptable charging current (PACC) is proposed on the basis of the polarization time factor: the CC charging time length is equal to the larger polarization time factor at a specified SOC point. If the external battery voltage exactly gets the preset cutoff voltage at the end of charging, then the charging current is the acceptable charging current value at this SOC point.

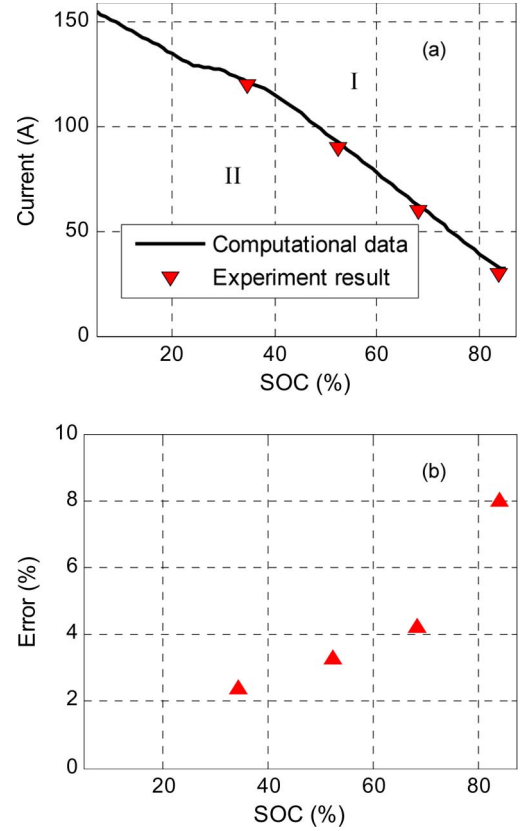


Fig. 10. (a) Acceptable charging current compared with experiment results and (b) calculated error.

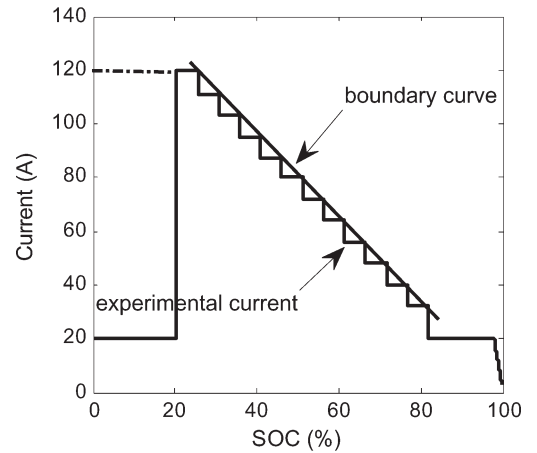


Fig. 11. Optimized charging strategy diagram.

The PACC relational expression is as follows:

$$V_{OCV}(SOC_0 + I\tau/C) + IR + V_P(I) = V_{cutoff}. \quad (8)$$

The iterative calculation PACC is shown in Fig. 10(a). The map is divided into two parts by PACC: (I) acceptable charging area under the curve and (II) unsuitable charging area. The battery can be charged in the region (I), which affects battery scarcely. The current will cause polarization increasing too large in the region (II), so that the battery cannot be fully charged. Here, select the larger time factor as charging time, because after such a long time, polarization voltage tends

TABLE II
COMPARISON OF PROPOSED CHARGING MODE AND CC–CV CHARGING MODE

Experiment	Charging mode	Total charging time /min	CV charging time /min	CV charging capacity /Ah	Equivalent rate /C	Average polarization voltage / mV	20%-80% SOC charging time /min
I	Proposed	105	7.2	1.082	0.57	79	33.6
II	0.5C CC–CV	128	9	3.023	0.47	80	72
III	1C CC–CV	69	10.2	4.514	0.87	102	36

to be stable, and jump over the dynamic change stage of polarization voltage. According to battery tests, parameters are set as follows: the cutoff voltage is 3.65 V; the ohmic internal resistance is 1.37 mΩ; the charging time is 1000 s. The result of acceptable charging current at different SOC is shown in Fig. 10. Compared to experimental data, the error is within 10%.

Considering the maximum charging current is 2C recommended by the battery manufacturer, the maximum charging current is therefore set as 2C. In addition, the battery polarization is large at the initial charging stage, in which small current charging is applied. The optimized charge strategy diagram is illustrated in Fig. 11. The battery is first charged at C/3 ranging from 0% SOC to 20% SOC. The current of 2C is applied after 20% SOC until it intersects with PACC. Then, every 5% SOC, the charging current is reduced according to the boundary curve. The charging current will be reduced to C/3 at around 80% SOC, followed by a C/3 CC–CV process.

Table II is the comparison between the proposed charging mode and the 1C and 0.5C CC–CV charging modes. It can be observed in Table II that the total charging time of experiment I is medium. CV charging time is almost equal, around 10 min. CV charging capacity of the three experiments increases in sequence. The initial CV charging currents of three experiments are C/3, 0.5C, and 1C. The polarization is increasing as the charge current increases. The actual equivalent charging rate of the proposed charging mode is medium between experiment II and experiment III. It can also be found in Table II that, compared with the 0.5C CC–CV charging mode (Experiment II), the charging time and equivalent charging rate of the proposed charging mode (Experiment I) are improved by 18% and 21.3%, respectively. The charging time and charging capacity at the CV charging stage are less than 0.5C CC–CV charging mode, having higher charging efficiency, requiring only CC charging in practice. In addition, the average polarization voltage of both charging modes is nearly the same. It is suggested that the proposed charging mode can well balance charging time and charging polarization compared to 0.5C CC–CV charging mode. Furthermore, the SOC ranging from 20% to 80% is chosen as the fast charging region in the proposed charging mode since it is the commonly used interval of the battery particularly for plug-in hybrid EVs. Compared with 1C CC–CV charging mode (Experiment III), the charging time is approximately identical from 20% to 80% SOC, and the average polarization voltage is much less than 1C CC–CV charging mode. From a temperature rise point of view, the charging current of the proposed charging mode is gradually decreasing, and the temperature rise is slowing down, which can effectively prevent the battery from thermal runaway.

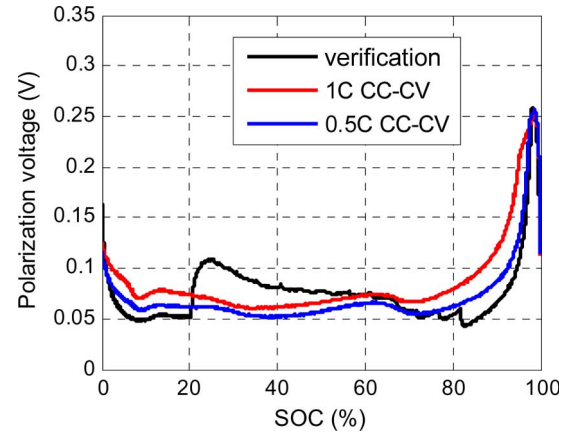


Fig. 12. Polarization voltage profiles.

Therefore, the proposed charging mode can balance charging speed, charging polarization, and temperature rise.

Fig. 12 is the comparison of polarization voltage profiles for the three experiments. In 0%–20% SOC and 80%–100% SOC, the polarization of experiment I is the lowest due to the low charging current. With charging current increasing suddenly, the polarization rises to the maximum at 20% SOC point. The polarization is going down as the current is falling. The minimum polarization of experiment I arrives at the 80% SOC.

The battery temperature distribution obtained by the thermal infrared imaging device at three charging modes is shown in Fig. 13. Fig. 13(a) is acquired at the maximum current of 2C of the proposed charging mode; Fig. 13(b) is obtained at the initial CV stage of the proposed charging mode. Fig. 13(c) and (d) are captured at the late charge stage of CC–CV, so that we can get the maximum temperature rise of every experiment. It is shown in Fig. 13 that 0.5C CC–CV charging mode has the minimum temperature range (298.15–303.15 K), and the 1C CC–CV picture has the maximum temperature rise (9.7 K) whereas the proposed charging mode is between the two. Furthermore, Fig. 13(b) shows that the temperature of battery drops when charging current decreases. The electrodes are the hottest due to high current density and contact resistance. The results indicate that temperature rise of the proposed charging mode is also acceptable.

All the experiments are done with two batteries that are randomly selected from ten batteries made from the same batch production. The results and methods have generality and representativeness and can be reproduced.

For modular Li-ion batteries, the equalization problem [22] and impedance measurement problem [23] are inevitable. The final acceptable charging current curve should take temperature rise, battery life, and charge efficiency into account for getting

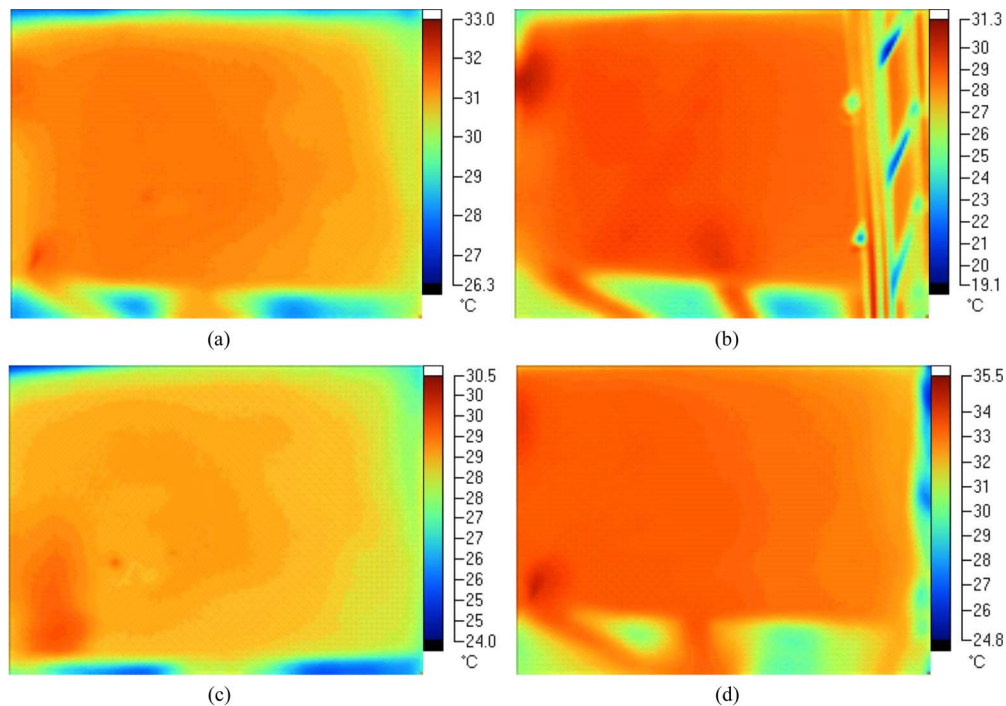


Fig. 13. Battery temperature distribution obtained by the thermal infrared imaging device at three charging modes. (a) Proposed charging mode at maximum current; (b) proposed charging mode at initial CV stage; (c) 0.5C CC-CV charging mode; (d) 1C CC-CV charging mode.

reasonable charging current values. In addition, an acceptable charging current curve may be not the practical applied current. The acceptable charging current taking multiple constraints into account requires further investigation.

IV. CONCLUSION

The analysis of polarization voltage features at different charging rate indicates that polarization voltage is high on both ends of SOC but low in the middle range of SOC, and the shape is similar to a bowl. In the middle range of SOC, an approximate linear relationship exists between the steady-state polarization voltage and the charging rate. The two time constants representing polarization voltage change are in 10- and 1000-s orders of magnitude, respectively, which correspond to three charging reaction processes. The dynamic polarization voltage exhibits a lagged effect and an overshoot effect when batteries are charged with varying current. Depending on polarization voltage characteristics, an acceptable charging current curve is proposed, taking polarization time constants into account, which can help design the practical charging strategy.

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